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What Is Next Generation Command and Control?

By Command Sgt. Maj. Waylon D. Petty

4th Infantry Division & Fort Carson

4th Infantry Division's Ties to Command and Control

So, no kidding, there I was ... a young sergeant M1A2 Abrams System Enhancement Package (SEP) gunner in C Company, 3rd Battalion, 67th Armor Regiment, 2nd Brigade Combat Team, 4th Infantry Division, out of Fort Hood, Texas.

On Jan. 3, 2003, President George W. Bush visited, and I was voluntold to escort him to one of the dining facilities for lunch.

After we ate, the President whispered to me, "Sgt. Petty, are your tanks ready?"

I won't tell you what I was thinking internally, but I responded, "They are Mr. President."

He then stood up and told everyone in the room, "Y'all go get 'em!"

In that moment, I had no idea who we were going to go get. That evening, 4ID received the order to deploy to Iraq, which became the start of Operation Iraqi Freedom.

I tell that story because, at that time, 4ID was known as the *Digital Division*, which was equipped with the Force XXI Battle Command Brigade and Below (FBCB2) system.

Force XXI was the Army's comprehensive modernization initiative to transform into a digitized, information-age force for the 21st century. That might

sound familiar. The problem with Force XXI was that it lacked modularity and agility.

Since its inception, various forms of wireless communication technology and the ability to transfer large amounts of data developed rapidly. Army leadership was aware of the need to invest in modernizing the Command and Control (C2) ecosystem to meet the Force XXI intent.

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Next Generation Command and Control (NGC2) will bring the latter through 4ID prototyping efforts, which embody a *Digital Culture*.

How It Started

Mid-July 2002, the entire division staff with brigade and battalion command teams met in a Fort Carson auditorium for an NGC2 introduction by Maj. Gen. Patrick J. Ellis, 4ID commanding general.

He displayed two slides. The first described the whole-stack approach broken down into four layers: Transport, Integration, Data, and Applications. Leaders in the room, including myself, tried to visualize what was presented and how it applied to what they know today.

To be frank, it was a lot to process because we were all used to the way we simplified command and control over the last two decades. This new approach was vastly different.

The second slide showed a number representing the days until Project Convergence Capstone 6 (PC-C6) was to launch, which meant most of the division would be in the box at the National Training Center validating NGC2.

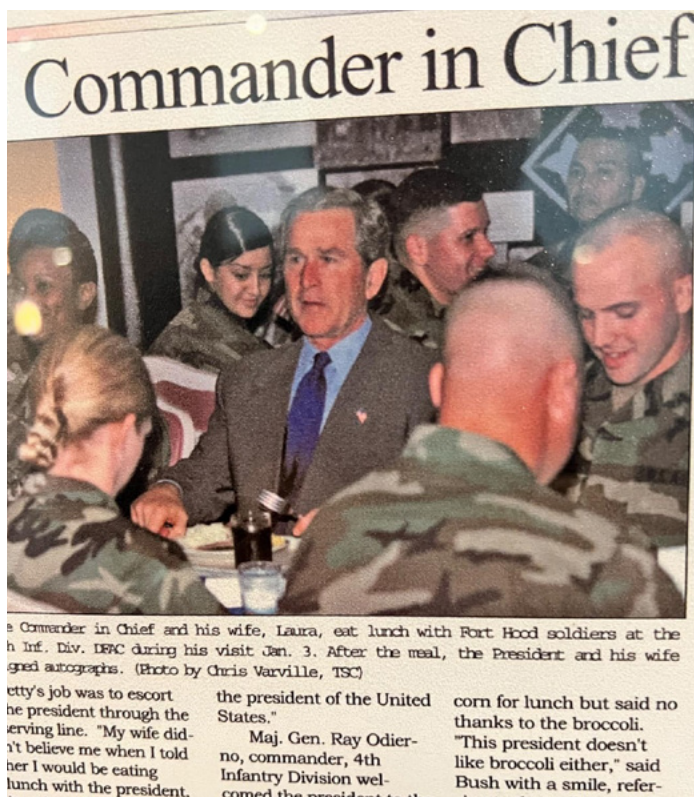
So, imagine being shown one slide you don't quite understand, fundamentally changing command and control with very little to no technology on Fort Carson yet, and being told you will fight using NGC2 in less than a year.

Fortunately, in Ellis' previous position as Network Cross Functional Team director, he worked for over a year to transition C2 Fix (an initiative streamlining and modernizing division and assigned BCTs' communications networks) into NGC2, which is what 4ID is prototyping for the Army today.

How We Prototype

Since we received that initial NGC2 overview brief I can tell you the entire division now speaks NGC2, not just the 4ID commanding general.

For two weeks every month, we run a series of mission threads to test warfighting systems within the NGC2 architecture. We call these NGC2-focused periods Ivy Stings.



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The first one was relatively basic, starting at the division level. Ivy Sting IV, the one we just completed in the first week of February, involved joint and multinational partners and the entire division down to the battalion level.

We are prototyping NGC2 but not sacrificing readiness and lethality. For the division, Ivy Stings are like mini-command post and warfighting exercises. Units are training teams and crews through company live-fire exercises.

In other words, NGC2 is not a separate training event; rather it is integrated into existing training schedules that build readiness across the division.

We're not doing this alone either. We have hundreds of industry and acquisition partners who live half their lives in Fort Carson. Industry partners and Soldiers work together on tasks that they need each other to test at scale; this is how you transform quickly and smartly.

Lock everyone in a room and nobody leaves until they meet their objectives while identifying new challenges to attack in the next turn.

However, none of this is perfect.

We face challenges every day to receiving authorization to operate on different layers, getting applications containerized into the architecture, or getting leadership to agree to move forward on a piece of tech, a common map, etc.

But that is exactly what prototyping is. If it doesn't work, throw it out. If it has enough potential, then we'll get it there.

What Does NGC2 Provide?

In short, NGC2 provides multiple options to communicate, the ability to pull/push more data, improved situational awareness, more efficient staff processes, and more effective rehearsals.

The architecture, if you will, for NGC2 is a whole-stack capability broken down into four layers: Transport, Integration, Data, and Applications. Employing and using these four layers will enable commanders, at echelon, to make faster, more informed decisions that will equate to lethality when and where we need it.

The Layers

The **transport layer** is the actual hardware that enables data movement, e.g., voice/audio, video feeds, common operating picture (COP), etc.

The **integration layer** takes the data and integrates it into the common data layer. For example, within the current C2 structure, data moves from radio to radio or from a satellite transportable terminal (STT) to an STT.

In NGC2, the integration layer provides multiple path options from a transport device to the data layer.

The first piece of tech most people think of is radios, and yes, that's part of it. At the company level, and from the boots-on-the-ground perspective, junior leaders will be equipped with a data-transmission device and/or an end-user device (EUD).

The data-transmission device will look like a dismantled radio, but it's capable of more than just transmitting/receiving voice communications; it's capable of operating across a wide range within the radio frequency portion of the electromagnetic spectrum.

Radio frequency includes technologies like 5G and Wi-Fi. The EUD will look and feel like a common cell phone, providing situational awareness integrated into the data layer and enabling functions like Voice over Internet Protocol (VoIP) to communicate with other teams.

There will be increasingly more transport and integration capability from the battalion level up. Such technologies will include satellite communication (SATCOM) such as proliferated low Earth orbit (pLEO), mid, and geostationary orbit (GEO) satellite constellations to provide low-latency with high-detail data to wide-coverage communications.

Additionally, data store at the edge will be integrated across echelons, which will enable dispersed C2 nodes to maintain connectivity in a Denied, Degraded, Intermittent, and Low-Bandwidth (DDIL) environment. Warfighting systems can continue to function while stationary, on the move, or even under the protection of

a building by connecting to an edge node.

Gone are the days of massive server stacks and huge network dishes that only push 10MB of data on a good day.

The **data layer** will use cloud and mesh network technology that provides two key attributes: 1) The ability to view, interact, and communicate across a COP without having to store massive amounts of data at the edge; and 2) Governs the kind and amount of data passed between nodes.

What does this mean?

Well, for starters, we won't have to burn Joint Battle Command-Platform (JBCP) bricks, which take hours for a small group of experts.



The Honorable Mike Obadal, Undersecretary of the Army, visits the 4th Infantry Division during Ivy Sting 5 for a demonstration of Next-Generation Command and Control capabilities. For the division, Ivy Stings are like mini command post and warfighting exercises. NGC2 is integrated into existing training schedules that build readiness across the division. (U.S. Army photo by Sgt. William Rogers)



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Additionally, the data layer COP is used by all, from squad leaders on the ground, Apache pilots in the air, to intelligence collectors holding custody of high-payoff targets.

The special ingredient is to seamlessly take classified data, convert it to Sensitive but Unclassified-Encrypted (SBU-E) data, and move it to the appropriate echelon to achieve the desired effect.

To me, the **application layer** has the most potential.

Think of the apps on your personal phone, where each app serves a purpose, executes a function, and receives updates when needed. These same principles apply to the NGC2 applications, where each warfighting system has multiple apps that serve specific warfighting functions operating on a common data layer.

The apps in the NGC2 architecture improve processes by automating tasks and functions, where possible. The technology exists today to track classes of supply, provide predictive logistics, produce a more efficient maintenance process, from PMCS to ordering parts, and tools to produce various types of orders faster.

Essentially, the potential applications will reduce the amount of time Soldiers collectively put into a process, which allows them to focus their energy elsewhere while reducing cognitive load.

You might be wondering if we're implementing artificial intelligence (AI) and/or machine learning (ML) into our applications.

Absolutely we are! I cannot speak on all of the specifics, but some of AI-enabled applications will drastically reduce operational battle rhythm meetings,

change the way we conduct rehearsals to make them more accurate/relevant, and create true shared understanding.

Imagine having an operational modeling tool that will tell you that Course of Action 3 is the one to go with and tells you why, based on current friendly/enemy capabilities, composition, and disposition.

Now that divisions are responsible for synchronizing intelligence, fires, and aviation, let's just say we have an App that uses an agentic AI-enabled fires workflow and it's scary good!

The So What

The Army chose 4ID to prototype NGC2 and to go fast — we feel that weight on our shoulders because we know we have to deliver.

4ID Soldiers live and breathe NGC2 every day because they owe that to the Army. That being said, I don't think the Army is going to wait for perfection — NGC2 will be coming soon to a theater near you.

Additionally, NGC2 will never end because this whole-stack approach was built to be modular and agile. It is not a physical object you're stuck with for the next 20 years; rather it is a blueprint to keep pace with modern technology.

This technology will give us C2 overmatch and enable lethality at the right place and at the right time. However, none of this technology matters if we're not ready to close with and destroy the enemy on the objective.

Discipline and fundamentals matter! Regardless of MOS, we all must be "Brilliant at the Basics." ■

Command Sgt. Maj. Waylon D. Petty is the 4th Infantry Division and Fort Carson, Colorado, command sergeant major. During his long career, he has served in a variety of leadership positions from tank gunner, tank commander, platoon sergeant, first sergeant, ROTC senior military instructor to command sergeant major. Petty previously served as the U.S. Army Armor School command sergeant major. He holds a Bachelor of Science degree in computer science from University of Maryland University College.



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